

data). In case the monitored machine is constituted as an intelligent master device, it could use the M2M device as a basic wireless modem. The M2M device would be fed data and then told to send it to the network. However, if the monitored machine is a mere bunch of switches and sensors, or is an intelligent slave device, the M2M device will then serve as the master device and regularly take inputs from the sensors and switches, or send data requests via the serial port.

Getting on to the cellular or satellite networks generally needs a gateway. A gateway receives the data from the wireless communication network and transforms it so that it can be sent to the network operation centre. Data security aspects, such as authentication and access control, are taken care of by the gateway and the application software.

Key M2M application areas

Today, M2M technologies have entered several walks of life. Energy, transport, real estate and agriculture are some major areas where this technology is fairly evolved. Benefits of M2M go beyond making life easier and smarter—they also include sustainable development of our planet.

Global warming is a serious environmental concern increasingly threatening our planet. The higher the CO₂ and other greenhouse gas (GHG) emissions, the greater is the global warming impact. Hence it is important that we cut down GHG emissions as much as possible, and M2M technologies help in achieving this goal too.

Energy. A smart meter is an electrical meter that records consumption of electric energy in intervals of an hour or less and transmits this data at least every day back to the utility central system for monitoring and billing purposes.

Hence smart meters enable two-way communication between the meter and the central system using M2M technology and form a part of advanced metering infrastructure. This leads to a smart grid which is a



A smart meter (Source: <http://smartgridtech.wordpress.com>)

modernised electrical grid that uses ICT to collect, interpret and act on information—such as demand pattern of consumers and supply pattern of the utility providers in an automated manner.

Time-of-day pricing, demand management, load balance and load optimisation help to improve the efficiency, reliability, economics and sustainability of the production and distribution of electricity.

M2M can also raise energy efficiency levels with regards to production and transmission, and hence further bring down CO₂ emissions by enabling usage of more renewable fuel resources.

It is estimated that M2M could save more than 2.0Gt of carbon dioxide emissions by 2020 in the energy sector alone by enabling the adoption of 'smart grid' technologies for both small and large-scale users, including smart meters and demand response systems.

Transport. In the transport sector, M2M applications range from the

simple fleet management of personal as well as commercial vehicles, maritime cargo shipping management to the more sophisticated surface transport GPS systems that adapt to varying road conditions (such as heavy road congestion ahead in the journey) to suggest substitute routes.

Route optimisation of personal and commercial vehicles results in cutting down unnecessary driving, less wastage of fuel and subsequently lower GHG emissions. BCG & GeSI estimate that M2M apps could save almost 1.9Gt of CO₂ emissions by 2020 in the transportation sector by optimising routes of planes, trains, trucks and ships, and ensuring that people and goods are moved as efficiently as possible.

In case of passenger vehicles, different ride-sharing and personal transport optimisation companies that use M2M are swiftly achieving credibility and market share, thus bringing down the total number of vehicle kilometres travelled.

Better fleet-management systems mean improved vehicle diagnostics, higher vehicle lifespan and lower operating expenditure. M2M can also help in inventory management to bring down losses by eliminating deadheading, for example, return journeys made by empty or underutilised vehicles.

Real estate. By using M2M technologies, it is possible to increase energy efficiency in the built environment without incurring capital expenditure and infrastructure changes.

M2M enables development of extensive and sophisticated building management systems, which monitor various parameters, such as the outside temperature and wind velocity.

These systems result in higher security and lower costs, for example, by switching lights and machines on and off automatically. Time is also saved as



Transport sector M2M apps often involve GPS monitoring (Source: <http://aggregate.tibbo.com>)